

Chapter 4

Properties of Pure Substance

The term **phase** refers to a quantity of matter that is homogeneous throughout space in both chemical composition and physical structure. Homogeneity in physical structure means that the matter is all solid, or liquid or all vapor (gas). A system can contain one or more phases. Two phases co-exist during the changes in phase called **vaporization**, **melting**, and **sublimation**. A **pure substance** is one that is uniform and invariable in chemical composition. A pure substance can exist in more than one phase, but its chemical composition must be the same in each phase.

4.1 Fixing the State

The *intensive* state of a closed system at *equilibrium* is its condition as described by the values of its intensive thermodynamic properties. From multiple observations, it has been repeatedly established that not all intensive thermodynamic properties are independent of each other. The intensive state can be uniquely determined by prescribing the values of a subset of *independent* thermodynamic intensive properties. Values for all other intensive thermodynamic properties are determined once this independent subset is specified. The **state principle**, a.k.a. the **state postulate**, tells us the number of independent intensive thermodynamic properties required to fix the intensive state. *From this point onwards, unless explicitly stated, the term “property” implies intensive thermodynamic property, and the term “state” implies an intensive state as defined by its intensive thermodynamic properties.* The *state principle* tells us that there is one independent property for each way a system’s energy can be varied independently. An independent property can be counted with heat transfer as one way of varying the energy, and another independent property can be counted for each relevant way the energy can be changed through work. Alternately, the number of independent properties required to fix the state is one plus the number of relevant work interactions. The term *simple system* is applied when there is only *one* way the system energy can be significantly altered by work as the system undergoes quasiequilibrium processes. Therefore, counting one independent property for heat transfer and another for the single work mode gives a total of two independent properties needed to fix the state of a simple system. **This is the state principle for simple systems.** The most common simple systems that we will encounter in this course are the simple *compressible* systems, where work done is $\int p dV$, and the energy transfer by work is associated with just the volume change, no other way. Other types of simple systems are simple *elastic* systems and simple *magnetic* systems.

Intensive properties such as velocity and elevation that are assigned values relative to datums *outside* the system are excluded from present considerations. Velocity and elevation are *mechanical* properties,

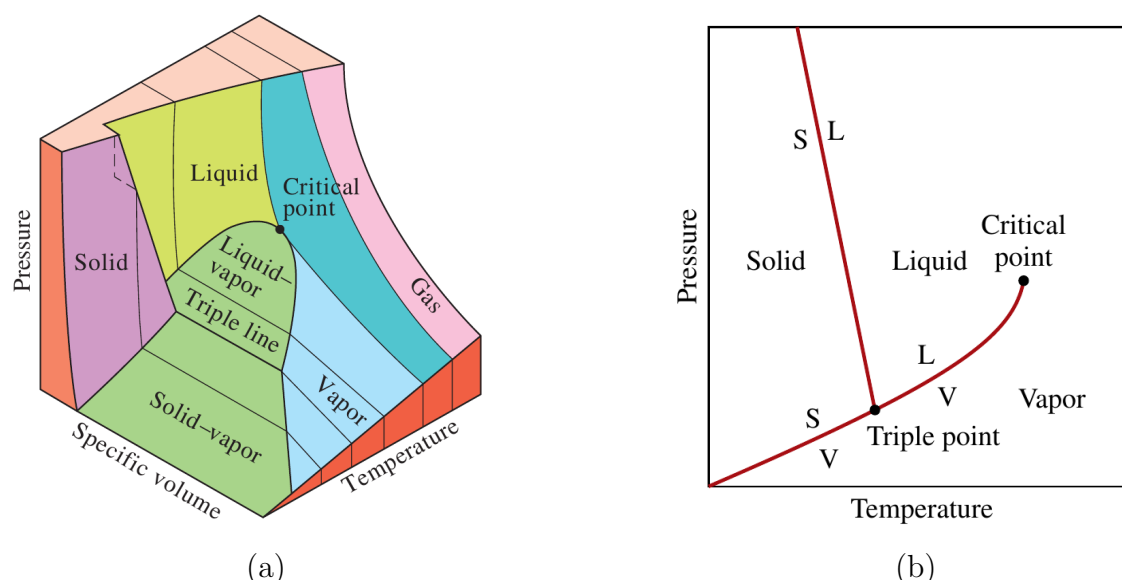


Figure 4.1: (a) Three-dimensional view of $p - v - T$ diagram of a substance that expands on freezing, (b) Projection $p - T$ diagram of the 3D view.

not thermodynamic properties. Velocity and elevation cannot be expressed as a function of any combination of thermodynamic intensive properties, nor can any thermodynamic intensive property be expressed in terms of velocity and elevation.

4.2 $p - v - T$ Relation

We begin our study of the properties of pure, simple compressible substances and the relations among these properties with pressure, specific volume, and temperature. From the state principle, we know that two properties are needed to fix the state of simple compressible systems. From experimental observations, it is known that specific volume (v) and temperature (T) can be regarded as two such independent properties and all other intensive properties, including pressure, can be written as functions of v and T , i.e., $p = p(v, T)$. If v and T are drawn as two orthogonal coordinate axes on the horizontal surface, then corresponding to each point in the (v, T) plane, we will have a unique value of pressure p . Let's represent pressure p on a vertical axis. Corresponding to each 2D point in the (v, T) plane there will be a 3D point in the 3D space (v, T, p) . If we join all these 3D points associated with all the 2D points in the (v, T) plane, we will obtain a 3D surface, which is called the $p - v - T$ surface.

Figure 4.1 illustrates a typical $p - v - T$ surface of a substance such as water that expands on freezing. The coordinates of a point on the $p - v - T$ surface represent the values that pressure, specific volume, and temperature would assume when the substance is at *equilibrium*.

There are regions on the $p - v - T$ surface labeled as *solid*, *liquid* and *vapor*. In these **single-phase** regions, the state is fixed by *any* two of the properties: pressure, specific volume, and temperature, any two of these of these properties are independent when single phase is present. Located between the single-phase regions are **two-phase regions**, where two phases exist in equilibrium: liquid-vapor, solid-liquid, and solid-vapor. Two phases can coexist during changes in phase such as vaporization, melting and sublimation. Within two-phase regions, pressure and temperature are not independent; one cannot be changed without changing the other. In these regions, the state cannot be fixed by

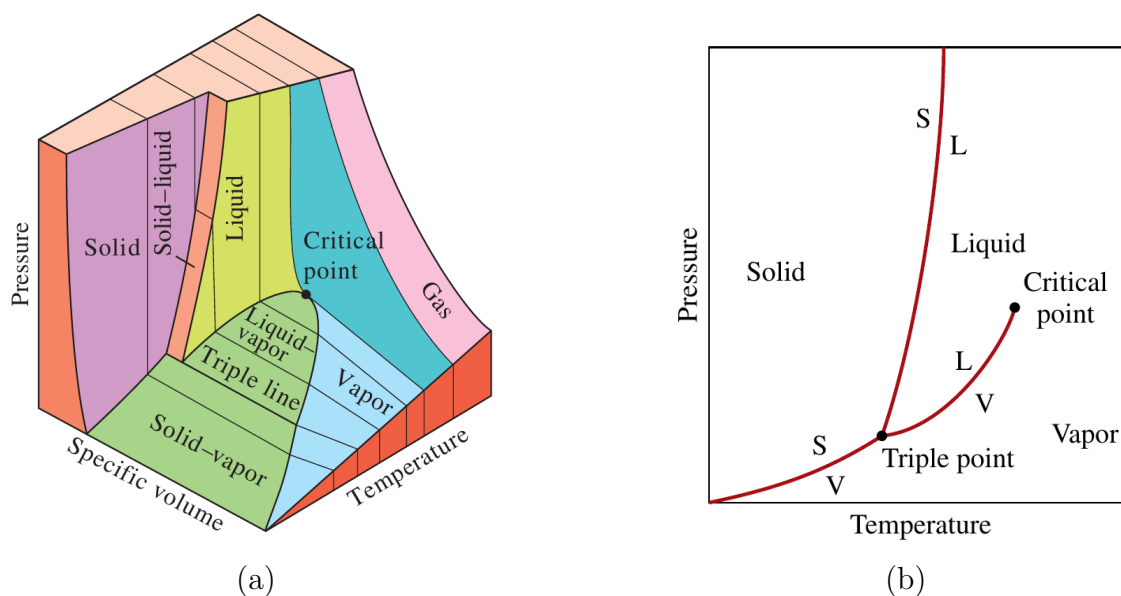


Figure 4.2: (a) Three-dimensional view of $p - v - T$ diagram of a substance that contracts on freezing, (b) Projection $p - T$ diagram of the 3D view.

temperature and pressure alone; however, the state can be fixed by specific volume and either pressure or temperature. Three phases can exist in equilibrium along the line labeled **triple line**.

A state at which a phase change begins or ends is called a **saturation state**. The dome-shaped region composed of the two-phase liquid-vapor states is called the **vapor dome**. The lines bordering the vapor dome are called saturated liquid and saturated vapor lines. At the top of the dome, where the saturated liquid and saturated vapor lines meet, is the **critical point**.

The *critical temperature* T_c of a pure substance is the maximum temperature at which liquid and vapor phases can coexist in equilibrium. The pressure at the critical point is called the *critical pressure* p_c . The specific volume at this state is the *critical specific volume*. At temperatures above the critical temperature, a substance exists as a vapor. However, this vapor cannot be liquified by increasing pressure. To liquify, the temperature will have to be brought down under the critical temperature.

Figure 4.2 displays $p - v - T$ diagram of a substance that contracts on freezing. Notice the slope of solid-liquid two-phase surface, and also the change in specific volume along an isotherm when phase changes from liquid to solid. The slope of the solid-liquid two-phase surface is dictated by the Clausius-Clapeyron equation, which we will derive in later chapters.

4.3 Projections of the $p - v - T$ surface

If the $p - v - T$ surface is projected onto the pressure-temperature plane, a property diagram known as a *phase diagram* results. As illustrated by Figure 4.1(b) and 4.2(b), when the surface is projected in this way, the two-phase *regions* reduce to *lines*. A point on any of these lines represents all two-phase mixtures at that particular temperature and pressure.

The term **saturation temperature** designates the temperature at which a phase change takes place at a given pressure, and this pressure is called the **saturation pressure** for the given temperature. It is apparent from the phase diagrams that for each saturation pressure there is a unique saturation

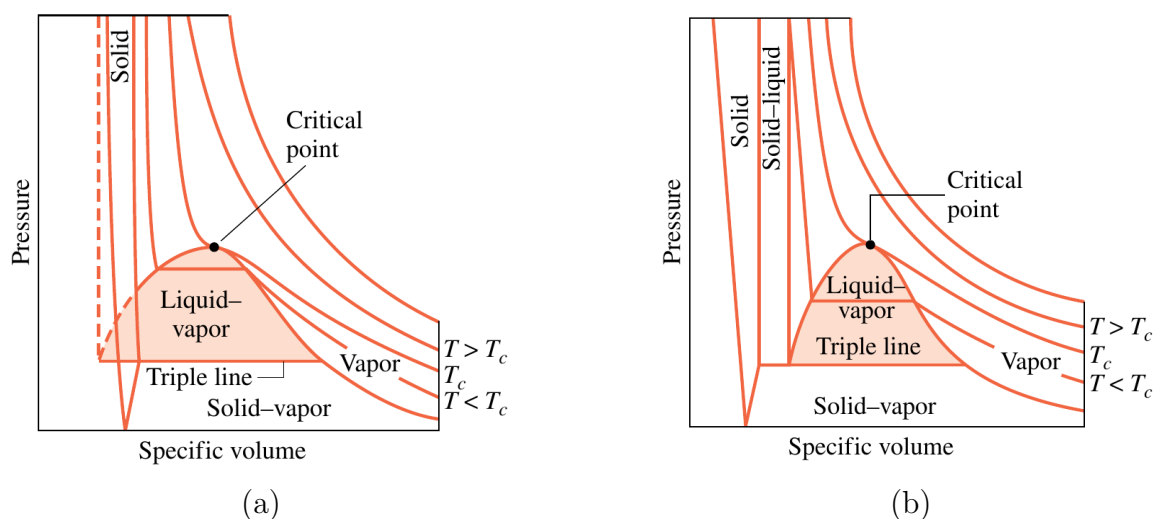


Figure 4.3: (a) $p-v$ diagram of a substance that expands on freezing, (b) $p-v$ diagram of a substance that contracts on freezing.

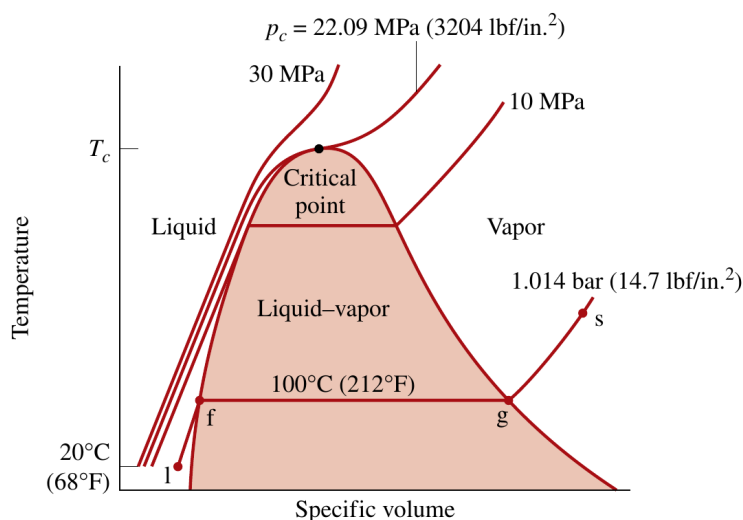


Figure 4.4: A temperature-specific volume diagram for water.

temperature, and conversely.

The triple *line* of the three-dimensional $p-v-T$ surface projects onto a *point* on the phase diagram. This is called the **triple point**. Recall that the triple point of water is used as a reference in defining temperature scales. The temperature *assigned* to triple point is 273.16 K. The *measured* pressure at the triple point of water is 0.6113 kPa.

Projecting the $p-v-T$ surface onto the pressure-specific volume plane results in a $p-v$ diagram, as shown in Figure 4.3. It can be seen that for any specified temperature *less than* the critical temperature, pressure remains constant as the two-phase liquid-vapor region is traversed, but in the single-phase liquid and vapor regions the pressure decreases at fixed temperature as specific volume increases. For temperature greater than or equal to the critical temperature, pressure decreases continuously at fixed temperature as specific volume increases. There is no passage through the two-phase liquid-vapor region. The critical isotherm passes through a point of inflection at the critical point and the slope is zero there.

A typical $T-v$ diagram (of water) is illustrated in Figure 4.4. To facilitate the use of such a

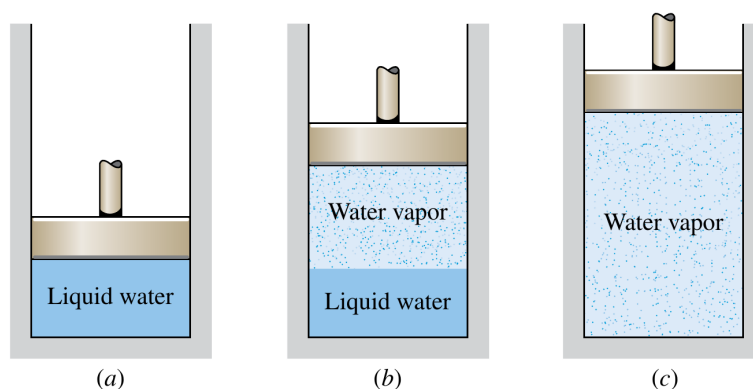


Figure 4.5: Illustrating constant pressure change from liquid to vapor.

sketch, note the appearance of constant-pressure lines (isobars). For pressures less than the critical pressure, the pressure remains constant with temperature as the two-phase region is traversed. In the single-phase liquid and vapor regions, the temperature increases at fixed pressure as the specific volume increases. For pressures greater than or equal to the critical pressure, temperature increases continuously at fixed pressure as the specific volume increases. There is no passage across the two-phase liquid-vapor region.

4.4 Phase Change Processes

Consider a closed system consisting of a unit mass, say 1 kg, of liquid water at 20°C contained with a piston-cylinder assembly as illustrated in Figure 4.5(a). This state is represented by point 1 in Figure 4.4. Suppose the water is slowly heated while its pressure is kept constant and uniform throughout at 1.014 bar. As the system is heated at constant pressure, the temperature increases considerably while the specific volume increases slightly. Eventually, the system is brought to the state represented by f in Figure 4.4. This is the **saturated liquid** state corresponding to the specified pressure. For water at 1.014 bar, the saturation temperature is 100°C (212°F). The liquid states along the 1-f are sometimes referred to as the **subcooled liquid** states because the state temperature here is less than the saturation temperature at the state pressure. The states between 1-f are also referred to as **compressed liquid** because the state pressure is higher than the saturation pressure corresponding to the state temperature. The names *liquid*, *subcooled liquid*, and *compressed liquid* are used interchangeably.

When the system is at the saturated liquid state (state f of Figure 4.4), additional heat transfer at fixed pressure results in the formation of vapor without any change in temperature but with a considerable increase in specific volume. As shown in Figure 4.5(b), the system would now consist of a two-phase liquid-vapor mixture. When a mixture of liquid and vapor exists in equilibrium, the liquid phase is a saturated liquid and the vapor phase is a saturated vapor. If the system is heated further until the last bit of liquid has vaporized, it is brought to point g in Figure 4.4, the saturated vapor state. The intervening **two-phase liquid-vapor mixture** states can be distinguished from one another by the **quality**, an intensive property. For a two-phase liquid-vapor mixture, the ratio of the mass of vapor present to the total mass of the mixture is its **quality**, x . In symbols,

$$x = \frac{m_{\text{vapor}}}{m_{\text{liquid}} + m_{\text{vapor}}} \quad (4.1)$$

The value of quality ranges from zero to unity. At saturated liquid states, $x = 0$, and at saturated

vapor states, $x = 1.0$. Although defined as a ratio, the quality is frequently given as a percentage. Similar parameters can be defined for two-phase solid-vapor and two-phase solid-liquid mixtures.

Returning to **Figures 4.4** and **4.5**, when the system is at a saturated vapor state (state g on **Figure 4.4**), further heating at fixed pressure results in increases in both temperature and specific volume. The condition of the system would now be as shown in **Figure 4.5(c)**. The state s on **Figure 4.4** is representative of the states that would be attained by further heating while keeping the pressure constant. A state such as s is often referred to as a **superheated vapor** state because the system would be at a temperature greater than the saturation temperature corresponding to the given pressure.

Consider next the same thought experiment at the other constant pressures labeled on **Figure 4.4**: 10 MPa, 22.09 MPa, and 30 MPa. The first of these pressures are less than the critical pressure of water, the second is the critical pressure, and the third is greater than the critical pressure. Let the system initially contain a liquid at 20°C. If it were heated slowly at 10 MPa, the vapor would form at a higher temperature than the isobar at 1.014 bar. In addition, there would be less increase in specific volume from saturated liquid to saturated vapor, as evidenced by narrowing of the vapor dome. The general behaviour, however, will remain the same.

Consider next the behavior of the system if it were heated at the critical pressure or higher. As seen by following critical isobar ($P = 22.09$ MPa), there would be no change in phase from liquid to vapor. At all states there would be only one phase. At states where pressure is greater than critical pressure, the terms *liquid* and *vapor* tend to lose their significance. Still, for ease of reference to such states, we use the term *liquid* when the temperature is less than the critical temperature and *vapor* when the temperature is greater than the critical temperature. Fluids at pressures greater than the critical pressure *or* temperatures greater than the critical temperature are called **supercritical** fluids. There are some useful characteristics of such fluids, such as superior transportiveness like gases, ability to move through porous media and the ability to dissolve solutes, just like liquids.

Phase change processes of melting and sublimation occur in a similar way. However, for engineering applications liquid vapor phase change are of primary interest due to their superior transportiveness properties. This enables them to carry energy (heat) from one location to another, e.g., in coolant applications.

4.5 Retrieving Thermodynamic Properties

Thermodynamic properties such as internal energy, specific volume, enthalpy, entropy etc. of pure substances at different states could be retrieved in various ways, including tables, graphs, equations, and computer software. The ability to locate states on property diagrams is an important related skill. Skillful use of tables and property diagrams is pre-requisite for the effective use of software to retrieve thermodynamic property data. **Using softwares as a blackbox is highly discouraged**. Doing so leads to wrong calculations and improper design of thermal equipments.

We will first learn to extract properties of superheated vapor state. As shown in **Figure 4.6**, the properties have been shown for each pressure separately, which correspond to isobars at those pressures. Since pressure and temperature are independent properties in the single-phase liquid and vapor regions, they can be used to “fix the state” in these regions. Thus, the properties are provided in **Figure 4.6** as functions of pressure and temperature. For each pressure listed, For each pressure listed, the values given in the superheated vapor table begin with the saturated vapor state and then proceed to higher temperatures. If the states encountered when solving problems often do not fall

TABLE A-6												
Superheated water												
T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$P = 0.01 \text{ MPa (45.81°C)}^*$					$P = 0.05 \text{ MPa (81.32°C)}$				$P = 0.10 \text{ MPa (99.61°C)}$			
Sat. [†]	14.670	2437.2	2583.9	8.1488	3.2403	2483.2	2645.2	7.5931	1.6941	2505.6	2675.0	7.3589
50	14.867	2443.3	2592.0	8.1741								
100	17.196	2515.5	2687.5	8.4489	3.4187	2511.5	2682.4	7.6953	1.6959	2506.2	2675.8	7.3611
150	19.513	2587.9	2783.0	8.6893	3.8897	2585.7	2780.2	7.9413	1.9367	2582.9	2776.6	7.6148
200	21.826	2661.4	2879.6	8.9049	4.3562	2660.0	2877.8	8.1592	2.1724	2658.2	2875.5	7.8356
250	24.136	2736.1	2977.5	9.1015	4.8206	2735.1	2976.2	8.3568	2.4062	2733.9	2974.5	8.0346
300	26.446	2812.3	3076.7	9.2827	5.2841	2811.6	3075.8	8.5387	2.6389	2810.7	3074.5	8.2172
400	31.063	2969.3	3280.0	9.6094	6.2094	2968.9	3279.3	8.8659	3.1027	2968.3	3278.6	8.5452
500	35.680	3132.9	3489.7	9.8998	7.1338	3132.6	3489.3	9.1566	3.5655	3132.2	3488.7	8.8362
600	40.296	3303.3	3706.3	10.1631	8.0577	3303.1	3706.0	9.4201	4.0279	3302.8	3705.6	9.0999
700	44.911	3480.8	3929.9	10.4056	8.9813	3480.6	3929.7	9.6626	4.4900	3480.4	3929.4	9.3424
800	49.527	3665.4	4160.6	10.6312	9.9047	3665.2	4160.4	9.8883	4.9519	3665.0	4160.2	9.5682
900	54.143	3856.9	4398.3	10.8429	10.8280	3856.8	4398.2	10.1000	5.4137	3856.7	4398.0	9.7800
1000	58.758	4055.3	4642.8	11.0429	11.7513	4055.2	4642.7	10.3000	5.8755	4055.0	4642.6	9.9800
1100	63.373	4260.0	4893.8	11.2326	12.6745	4259.9	4893.7	10.4897	6.3372	4259.8	4893.6	10.1698
1200	67.989	4470.9	5150.8	11.4132	13.5977	4470.8	5150.7	10.6704	6.7988	4470.7	5150.6	10.3504
1300	72.604	4687.4	5413.4	11.5857	14.5209	4687.3	5413.3	10.8429	7.2605	4687.2	5413.3	10.5229

Figure 4.6: Properties of superheated water vapor.

exactly on the grid of values provided in the table, linear interpolation between adjacent values is done.

We will now show how to extract the properties of saturated vapor-liquid mixtures. An excerpt from the saturated table of water is illustrated in Figure 4.7. In the saturation mixture, some mass of matter is present as saturated liquid and the rest is present as saturated vapor. Along an isobar, all states in the saturation mixture have the same temperature and pressure. This is because there is a dependency between temperature and pressure during saturation. To distinguish between two saturation states, we need another variable, which can be, for numerical convenience, the **quality** of the mixture. In the table shown in Figure 4.7, the properties of saturated liquid and saturated vapor are provided. The property of the saturation mixture can be computed using these properties and quality as follows. Suppose we are interested in computing specific internal energy u of a saturated mixture having quality x . Let us suppose the mass of liquid is m_f and the mass of vapor is m_g . The total quantity of internal energy for the mixture is given by:

$$U = m_f \cdot u_f + m_g \cdot u_g \quad (4.2)$$

If the entire equation is divided by the total mass of the mixture $m_f + m_g$, we obtain:

$$u = (1 - x) \cdot u_f + x \cdot u_g \quad (4.3)$$

In fact, any intensive property q can be written in terms of quality x :

$$q = (1 - x) \cdot q_f + x \cdot q_g \quad (4.4)$$

Figure 4.8 shows the **pressure table**, because pressures are listed in the first column in convenient increments. The corresponding saturation temperatures are given in the second column. The **temperature table** shown in Figure 4.7 also lists the same data, only increments are in temperature. If a $p - T$ curve is made from the pressure table and another from the temperature table, those two curves will exactly overlap each other.

The thermodynamic tables don't present an exhaustive table for **supercooled liquid states**. Properties of supercooled liquids are not available especially at lower pressures. Approximate values of

TABLE A-4

Saturated water—Temperature table

Temp., T °C	Specific volume, m^3/kg			Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
	Sat. Press., P_{sat} kPa	Sat. liquid, ν_f	Sat. vapor, ν_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661
25	3.1698	0.001003	43.340	104.83	2304.3	2409.1	104.83	2441.7	2546.5	0.3672	8.1895	8.5567
30	4.2469	0.001004	32.879	125.73	2290.2	2415.9	125.74	2429.8	2555.6	0.4368	8.0152	8.4520
35	5.6291	0.001006	25.205	146.63	2276.0	2422.7	146.64	2417.9	2564.6	0.5051	7.8466	8.3517
40	7.3851	0.001008	19.515	167.53	2261.9	2429.4	167.53	2406.0	2573.5	0.5724	7.6832	8.2556
45	9.5953	0.001010	15.251	188.43	2247.7	2436.1	188.44	2394.0	2582.4	0.6386	7.5247	8.1633
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898
60	19.947	0.001017	7.6670	251.16	2204.7	2455.9	251.18	2357.7	2608.8	0.8313	7.0769	7.9082
65	25.043	0.001020	6.1935	272.09	2190.3	2462.4	272.12	2345.4	2617.5	0.8937	6.9360	7.8296
70	31.202	0.001023	5.0396	293.04	2175.8	2468.9	293.07	2333.0	2626.1	0.9551	6.7989	7.7540
75	38.597	0.001026	4.1291	313.99	2161.3	2475.3	314.03	2320.6	2634.6	1.0158	6.6655	7.6812
80	47.416	0.001029	3.4053	334.97	2146.6	2481.6	335.02	2308.0	2643.0	1.0756	6.5355	7.6111
85	57.868	0.001032	2.8261	355.96	2131.9	2487.8	356.02	2295.3	2651.4	1.1346	6.4089	7.5435
90	70.183	0.001036	2.3593	376.97	2117.0	2494.0	377.04	2282.5	2659.6	1.1929	6.2853	7.4782
95	84.609	0.001040	1.9808	398.00	2102.0	2500.1	398.09	2269.6	2667.6	1.2504	6.1647	7.4151
100	101.42	0.001043	1.6720	419.06	2087.0	2506.0	419.17	2256.4	2675.6	1.3072	6.0470	7.3542
105	120.90	0.001047	1.4186	440.15	2071.8	2511.9	440.28	2243.1	2683.4	1.3634	5.9319	7.2952
110	143.38	0.001052	1.2094	461.27	2056.4	2517.7	461.42	2229.7	2691.1	1.4188	5.8193	7.2382
115	169.18	0.001056	1.0360	482.42	2040.9	2523.3	482.59	2216.0	2698.6	1.4737	5.7092	7.1829
120	198.67	0.001060	0.89133	503.60	2025.3	2528.9	503.81	2202.1	2706.0	1.5279	5.6013	7.1292

Figure 4.7: Properties of saturated mixture of liquid water and vapor - Temperature table.**TABLE A-5**

Saturated water—Pressure table

Press., P kPa	Specific volume, m^3/kg			Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
	Sat. temp., T_{sat} °C	Sat. liquid, ν_f	Sat. vapor, ν_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
1.0	6.97	0.001000	129.19	29.302	2355.2	2384.5	29.303	2484.4	2513.7	0.1059	8.8690	8.9749
1.5	13.02	0.001001	87.964	54.686	2338.1	2392.8	54.688	2470.1	2524.7	0.1956	8.6314	8.8270
2.0	17.50	0.001001	66.990	73.431	2325.5	2398.9	73.433	2459.5	2532.9	0.2606	8.4621	8.7227
2.5	21.08	0.001002	54.242	88.422	2315.4	2403.8	88.424	2451.0	2539.4	0.3118	8.3302	8.6421
3.0	24.08	0.001003	45.654	100.98	2306.9	2407.9	100.98	2443.9	2544.8	0.3543	8.2222	8.5765
4.0	28.96	0.001004	34.791	121.39	2293.1	2414.5	121.39	2432.3	2553.7	0.4224	8.0510	8.4734
5.0	32.87	0.001005	28.185	137.75	2282.1	2419.8	137.75	2423.0	2560.7	0.4762	7.9176	8.3938
7.5	40.29	0.001008	19.233	168.74	2261.1	2429.8	168.75	2405.3	2574.0	0.5763	7.6738	8.2501
10	45.81	0.001010	14.670	191.79	2245.4	2437.2	191.81	2392.1	2583.9	0.6492	7.4996	8.1488
15	53.97	0.001014	10.020	225.93	2222.1	2448.0	225.94	2372.3	2598.3	0.7549	7.2522	8.0071
20	60.06	0.001017	7.6481	251.40	2204.6	2456.0	251.42	2357.5	2608.9	0.8320	7.0752	7.9073
25	64.96	0.001020	6.2034	271.93	2190.4	2462.4	271.96	2345.5	2617.5	0.8932	6.9370	7.8302
30	69.09	0.001022	5.2287	289.24	2178.5	2467.7	289.27	2335.3	2624.6	0.9441	6.8234	7.7675
40	75.86	0.001026	3.9933	317.58	2158.8	2476.3	317.62	2318.4	2636.1	1.0261	6.6430	7.6691
50	81.32	0.001030	3.2403	340.49	2142.7	2483.2	340.54	2304.7	2645.2	1.0912	6.5019	7.5931
75	91.76	0.001037	2.2172	384.36	2111.8	2496.1	384.44	2278.0	2662.4	1.2132	6.2426	7.4558
100	99.61	0.001043	1.6941	417.40	2088.2	2505.6	417.51	2257.5	2675.0	1.3028	6.0562	7.3589
101.325	99.97	0.001043	1.6734	418.95	2087.0	2506.0	419.06	2256.5	2675.6	1.3069	6.0476	7.3545
125	105.97	0.001048	1.3750	444.23	2068.8	2513.0	444.36	2240.6	2684.9	1.3741	5.9100	7.2841
150	111.35	0.001053	1.1594	466.97	2052.3	2519.2	467.13	2226.0	2693.1	1.4337	5.7894	7.2231

Figure 4.8: Properties of saturated mixture of liquid water and vapor - Pressure table.

u , v , h and s of supercooled liquid can still be computed using saturated liquid data. It has been observed that for liquids specific volume and specific internal energy change very little with pressure at a fixed temperature. Because the values of v and u vary only gradually as pressure changes at fixed temperature, the following approximations are reasonable.

$$v(T, p) \approx v_f(T) \quad (4.5)$$

$$u(T, p) \approx u_f(T) \quad (4.6)$$

That is, for liquids v and u may be evaluated at the saturated liquid state corresponding to the temperature at the given state. An approximate value of h at liquid states can be obtained by using the above approximations and the definition $h = u + pv$:

$$h(T, p) \approx u_f(T) + p v_f(T) \quad (4.7)$$

$$h(T, p) \approx h_f(T) + v_f(T) [p - p_{\text{sat.}}(T)] \quad (4.8)$$

4.6 Derivative Properties

Derivatives of a thermodynamic property with respect to other thermodynamic properties, sometimes also called **secondary properties**, are themselves thermodynamic properties. For example, consider the specific volume v as a function of temperature T and pressure p , which we express as $v(T, p)$. The **total differential** of v is given by the chain rule of calculus:

$$dv(T, p) = \left(\frac{\partial v}{\partial T} \right)_p dT + \left(\frac{\partial v}{\partial p} \right)_T dp \quad (4.9)$$

The derivative $\left(\frac{\partial v}{\partial T} \right)_p$ is the slope of an isobar on a $v - T$ plot. Similarly, $\left(\frac{\partial v}{\partial p} \right)_T$ is the slope of an isotherm on a $v - p$ plot. These partial derivatives are used to define compressibilities of a substance as follows. The **isobaric compressibility** is defined as:

$$\beta \equiv \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p \quad (4.10)$$

and measures the percentage volume increase per unit of temperature increase at constant pressure. The **isothermal compressibility** is defined as:

$$\kappa \equiv -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T \quad (4.11)$$

and measures the percentage volume increase per unit of pressure increase at constant temperature.

A similar exercise can be done with the specific internal energy u . Thinking of $u(T, v)$ as a function of temperature and specific volume, the total differential du is:

$$du = \left(\frac{\partial u}{\partial T} \right)_v dT + \left(\frac{\partial u}{\partial v} \right)_T dv \quad (4.12)$$

The first of these derivatives is defined to be c_v ,

$$c_v \equiv \left(\frac{\partial u}{\partial T} \right)_v \quad (4.13)$$

Consider the heating of a substance maintained at constant volume by a rigid container. An energy balance on the substance gives:

$$\delta q = du + p \cancel{dv}^0 \quad (4.14)$$

$$\delta q = du \quad (4.15)$$

$$\delta q = c_v dT \quad (4.16)$$

Here we have used the fact that for a constant-volume process, $u = u(T)$ instead of $u(T, v)$ and hence $c_v = du/dT$. Note that c_v is the energy input as heat, per unit mass, per unit of temperature rise, for a constant-volume heating process; therefore, c_v is called the **specific heat at constant volume**, or *isochoric* specific heat. In general $c_v = c_v(T, v)$, but we can often treat c_v as a constant in engineering problems, especially over limited ranges of T and v .

Now consider the heating of substance maintained at constant pressure in a container with a weighted piston on the top as shown in **Figure 4.9**.

An energy balance on the substance gives:

$$\delta q = du + \delta w \quad (4.17)$$

$$\delta q = du + p dv \quad (4.18)$$

For a constant-pressure process, $\delta q = d(u + pv) = dh$. Thinking of enthalpy $h = h(T, p)$, the total differential dh is:

$$dh = \left(\frac{\partial h}{\partial T} \right)_p dT + \left(\frac{\partial h}{\partial p} \right)_T dp \quad (4.19)$$

The first of these derivatives is denoted by c_p ,

$$c_p \equiv \left(\frac{\partial h}{\partial T} \right)_p \quad (4.20)$$

Now, since $dp = 0$ for this system, the energy balance yields

$$\delta q = c_p dT \quad (4.21)$$

Here we have used the fact that for a constant pressure process, $h = h(T)$ instead of $h(T, p)$, so $c_p = dh/dT$. Note that the c_p is the energy transfer as heat, per unit mass, per unit of temperature rise, or a constant-pressure heating process; therefore, c_p is called the **specific heat at constant pressure**, or *isobaric* specific heat. In general $c_p = c_p(T, p)$, but we can often treat c_p as a constant in engineering applications, especially over limited ranges of T and p . If you look at $p - h$ chart of pure fluid, you see that inside the vapor-liquid dome, T does not change at constant p , but h does, and hence c_p is infinite.

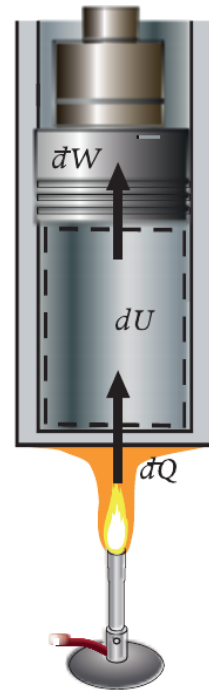


Figure 4.9: Constant pressure heating.

4.7 Enthalpy

For reasons that will become clear in the next chapter, a property related to the internal energy usually appears on thermodynamic property charts instead of (or in addition to) u . The *enthalpy* is defined as the property combination

$$H \equiv U + pV \quad (4.22)$$

Addition pV to U makes sense since both have units of energy. The related intensive property is the specific enthalpy $h \equiv H/M$,

$$h \equiv u + pv \quad (4.23)$$

If one can find h , p , and v in a chart or table, u is easily determined, if needed, by subtracting pv . Since, h occurs more naturally in energy balances on *open systems* (control volumes), engineers decided a long time ago that it would be better to get or visualize h easily rather than u , and hence most tables and charts display h information but not u information.

4.8 Incompressible substance

There are regions in the phase diagram, where the specific volume of liquid water varies little and the specific internal energy varies mainly with temperature. The same general behavior is exhibited by the liquid phases of other substances and by solids. The approximations used for computing properties of subcooled liquids are based on these observations.

To simplify evaluations involving liquids or solids, the specific volume (density) is often assumed to be constant and the specific internal energy assumed to vary only with temperature. A substance idealized in this way is called **incompressible**. Thus for a substance modeled as incompressible,

$$v = \text{constant} \quad (4.24)$$

$$u = u(T) \quad (4.25)$$

Since the specific internal energy of a substance modeled as incompressible depends only on temperature, the specific heat c_v is also a function of temperature alone:

$$c_v(T) = \frac{du}{dT} \quad (\text{incompressible}) \quad (4.26)$$

This is expressed in ordinary derivative because u depends only on T . Although, the specific volume is constant and internal energy depends on temperature only, enthalpy varies with both pressure and temperature according to:

$$h(T, p) = u(T) + pv \quad (\text{incompressible}) \quad (4.27)$$

If we take the partial derivative of the **Equation (4.27)** against T at constant pressure, we obtain:

$$\left(\frac{\partial h}{\partial T} \right)_p = \frac{du}{dT} + p \left(\frac{\partial v}{\partial T} \right)_p \quad (4.28)$$

$$\text{or,} \quad (4.29)$$

$$c_p = \frac{du}{dT} + p v \beta \quad (4.30)$$

$$c_p = \frac{du}{dT} \quad (4.31)$$

For an incompressible substance both β and κ defined in **Equations (4.10)** and **(4.11)**, respectively, are zero. Therefore, the second term on the right in **Equation (4.30)** can be dropped. Since c_p in **Equation (4.31)** is a regular derivative of $u(T)$, $c_p = c_p(T)$. Comparing **Equations (4.26)** and **(4.31)**, we find that for incompressible substances, $c_p = c_v = c(T)$.

The changes in specific internal energy and specific enthalpy between two states are given, respectively, by

$$u_2 - u_1 = \int_{T_1}^{T_2} c(T) dT \quad (\text{incompressible}) \quad (4.32)$$

$$\begin{aligned} h_2 - h_1 &= u_2 - u_1 + v(p_2 - p_1) \\ &= \int_{T_1}^{T_2} c(T) dT + v(p_2 - p_1) \quad (\text{incompressible}) \end{aligned} \quad (4.33)$$

4.9 Ideal Gas

An **ideal** or perfect gas is any substance that obeys the equation of state:

$$p = \rho R T \quad (4.34)$$

Here, $R = \hat{R}/\hat{M}$, where $\hat{R} = 8.3143 \text{ J}/(\text{mol}\cdot\text{K})$ is the *universal gas constant* and $\hat{M}$ is the molar mass (mass per mole) of the substance. Equivalent form is:

$$p v = R T \quad (4.35)$$

These are the intensive forms of the defining equation. For a sample of mass M containing N moles in a volume V , the extensive forms are:

$$p V = M R T \quad (4.36)$$

and

$$p V = N \hat{R} T \quad (4.37)$$

One can tell where the ideal gas model is a good approximation by looking at the thermodynamic property charts. For example, on **Figure 4.10**, the line for $p = 0.1 \text{ MPa}$ is very nearly straight throughout the gas region, and displaced in volume at any given temperature by a factor of 10 from the line for $p = 1 \text{ MPa}$, indicating that the ideal gas model would be a good approximation in that region. You can see that the ideal gas model is best at very high volumes, or low densities, when the molecules are on average very far apart. In the limit $\rho \rightarrow 0$, all substances behave like an ideal gas.

4.9.1 Internal energy of an ideal gas

The internal energy of the gas is the sum of the translational, rotational and vibrational energy of each molecule; there is no significant energy associated with intermolecular forces because the molecules are on average very far apart. As a result, the **internal energy of an ideal gas depends only upon temperature**, which is a function of the average molecular energy:

$$u = u(T) \quad (4.38)$$

If **Figure 4.11** is examined, one will find that at low pressures outside of the vapor dome the temperature lines and internal energy lines coincide, indicating that the internal energy is indeed independent of the density in the region where we expect ideal gas behaviour.

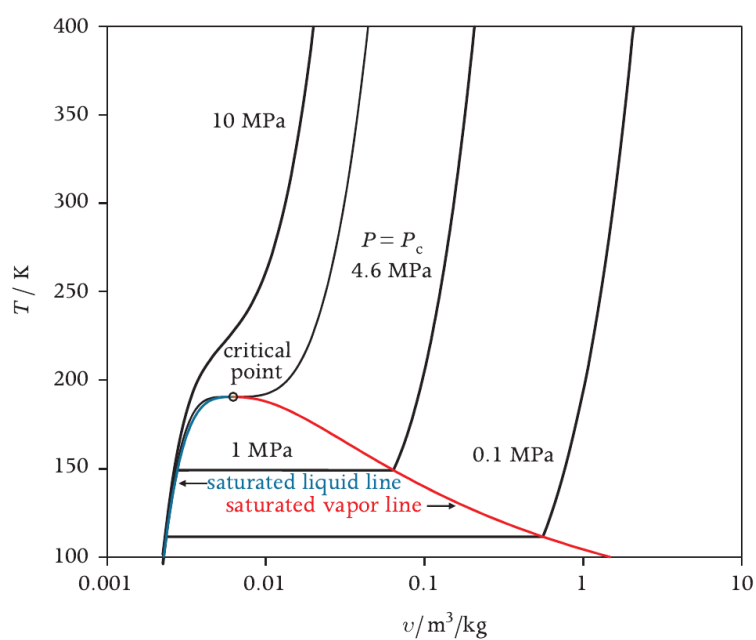


Figure 4.10: $T - v$ diagram for methane.

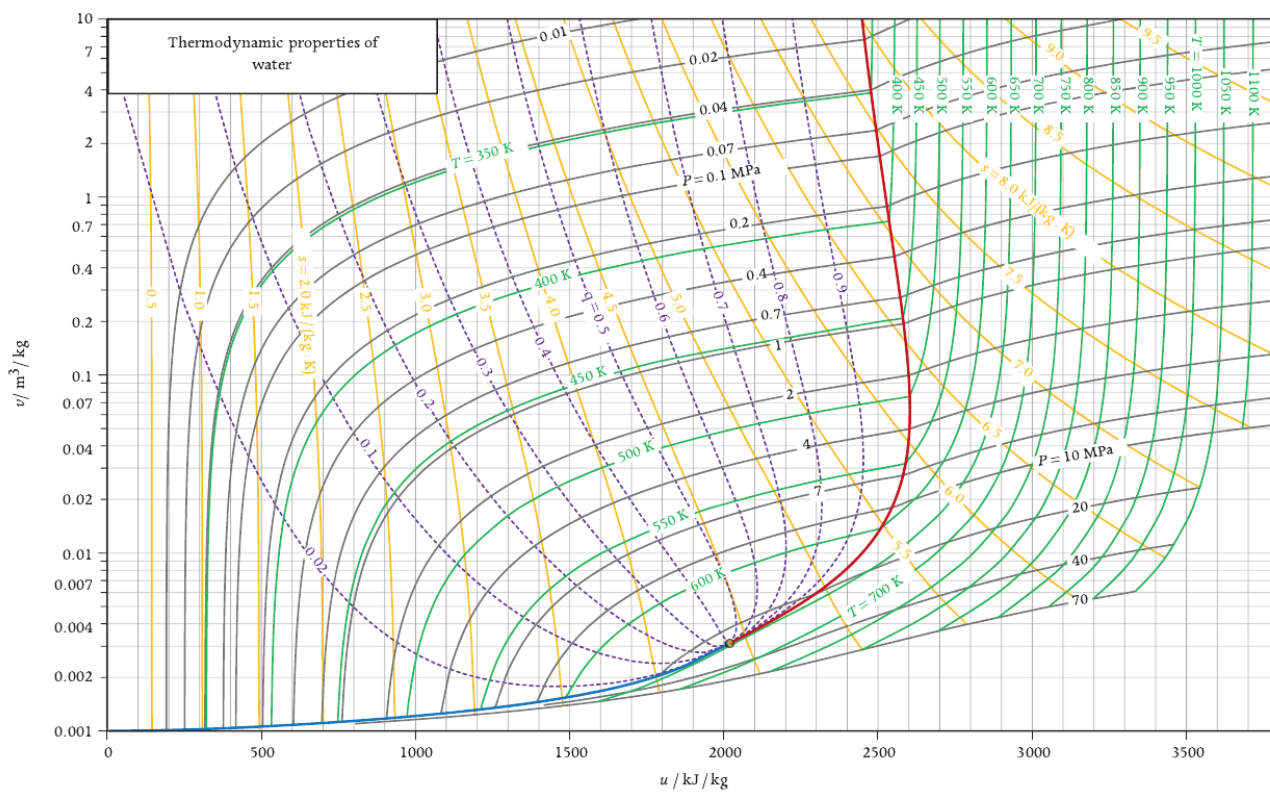


Figure 4.11: $v - u$ diagram of water.

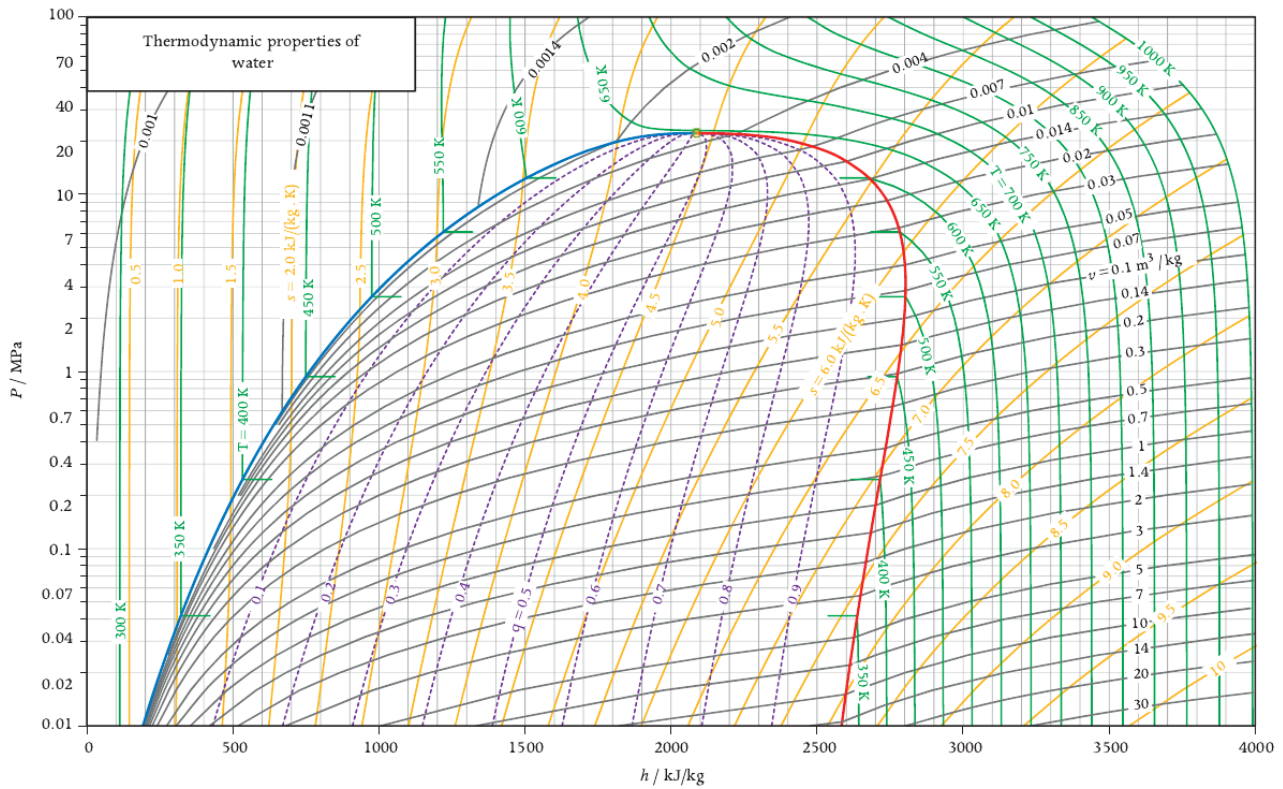


Figure 4.12: $p - h$ diagram for water.

4.9.2 Enthalpy for an ideal gas

Since $u = u(T)$ and $Pv = RT$, the enthalpy of an ideal gas $h = u + Pv$ is also only a function of temperature,

$$h = h(T) \quad (4.39)$$

Examination of the $p - h$ chart in Figure 4.12 confirms that at high volumes in the gas region the h lines coincide with T lines, indicating ideal gas behavior.

4.9.3 Specific heats for an ideal gas

Since u and h depend only upon temperature, so must the specific heats, hence

$$c_v = \left(\frac{\partial u}{\partial T} \right)_v = \frac{du}{dT} \quad \text{since } u = u(T) \quad (4.40)$$

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p = \frac{dh}{dT} = \frac{du}{dT} + R = c_v + R \quad (4.41)$$

For monoatomic gases the internal energy is due entirely to molecular translation, and translational energy is proportional to temperature. This means that the specific heats for monoatomic gases are constants. For more complex molecules the amount of energy associated with rotation and vibration increases with temperature in a nonlinear way, and hence the specific heats increase somewhat with temperature. However, over limited temperature ranges we often make the approximation that the specific heats are constant, which allows neat analytical solutions to many interesting problems.

Air, although a mixture of gases, can usually be treated as an ideal gas.

4.9.4 Pressure in an ideal gas

Here we will perform microscopic analysis to compute quantities like pressure, internal energy, specific heats etc. of an ideal gas. We start with pressure. The pressure in an ideal gas is determined by the rate of momentum transfer per unit area as the molecules collide with the container boundary. The molecules move at different at different speeds and in all possible directions. In order to compute the momentum transfer, let's first consider those particles moving such that their velocity component normal to the boundary is $\pm V_x$, see [Figure 4.13](#). In a small time interval dt , the number of these particles that strike a unit area of the boundary will be half of those contained within striking distance of the boundary, or

$$\frac{1}{2} \times n' \times V_x dt \quad (4.42)$$

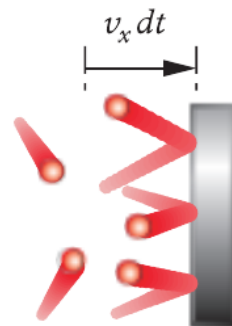


Figure 4.13: Determining the collision rate.

where n' is the number of molecules per unit volume that have this particular x -speed. The factor $1/2$ is needed because only half of the particles are moving towards the boundary. The other half is moving away from the boundary at the same x -component of velocity of V_x . Assume that these particles collide and rebound with the same x -speed in the opposite direction. Therefore, the x -component of momentum change in molecules is:

$$\frac{1}{2} n' V_x dt \times m (-V_x - V_x) = m n' V_x^2 dt \quad (4.43)$$

where m is the mass of a molecule. Now we add up the momentum transfers from molecules of all speeds, and divide by dt to get the rate of momentum transfer per unit area, i.e., the pressure

$$p = m \sum_{\text{speeds}} n' V_x^2 = m n \overline{V_x^2} \quad (4.44)$$

where n is the total number of particles per unit volume and $\overline{V_x^2} = \sum (n'/n) V_x^2$ is the mean-square velocity in the x -direction, averaged over all molecules. The total mean-square speed is

$$\overline{V^2} = \overline{V_x^2} + \overline{V_y^2} + \overline{V_z^2} \quad (4.45)$$

where $\overline{V_y^2}$ and $\overline{V_z^2}$ are the mean-square velocities in the y and z directions, respectively. Since the molecular motions are isotropic,

$$\overline{V_x^2} = \overline{V_y^2} = \overline{V_z^2} \quad (4.46)$$

Now, [Equation \(4.44\)](#) gives

$$p = \frac{1}{3} m n \overline{V^2} \quad (4.47)$$

Comparing [Equation \(4.47\)](#) with [Equation \(4.34\)](#), and noting that $\rho = m n$, we find that

$$RT = \frac{1}{3} \overline{V^2} \quad (4.48)$$

Thus we see that the temperature of an ideal gas provides a measure of its mean-square molecular translational velocity. This same expression holds for the translational energy of a polyatomic ideal gas.

4.9.5 Internal energy, enthalpy and specific heats of a monatomic ideal gas

If the gas is monatomic, its internal energy is entirely due to the translational kinetic energy of its atoms. The kinetic energy per unit mass is

$$u = \frac{1}{2} \overline{V^2} \quad (4.49)$$

Using Equation (4.48), the internal energy per unit mass (u) are found to be:

$$u = \frac{3}{2} R T \quad (4.50)$$

Note that the internal energy of the gas is only a function of its temperature, as discussed above, and that the internal energy per mole is the same for all monatomic ideal gases.

Since $h = u + pv$, the enthalpy for a monatomic gas,

$$h = \frac{5}{2} R T \quad (4.51)$$

Note that the enthalpy of the gas is only a function of its temperature, as discussed above, and that the enthalpy per mole is the same for all monatomic ideal gases.

Utilizing Equations (4.40) and (4.41), the specific heats, and specific heat ratio for an ideal monatomic gas are:

$$c_v = \frac{3}{2} R, \quad c_p = \frac{5}{2} R, \quad \gamma = \frac{c_p}{c_v} = \frac{5}{3} \quad (4.52)$$

4.9.6 Equipartition model

A simple but useful model for polyatomic gases is based on the assumption that each available mode of motion freedom contributes the same amount to the energy of the system. A monatomic molecule has three modes of freedom (translational in x , y and z directions) and hence from Equation (4.50), the energy per degree of freedom is

$$u = \frac{1}{2} R T \quad (4.53)$$

This simple rule is useful in estimating the internal energy, enthalpy, and specific heats of more complex molecules, as shown in the following.

4.9.7 Diatomic molecules

A rigid diatomic molecules has these three translational modes of freedom and two additional degrees of rotational freedom, for a total of five modes. One rotational degree of freedom's contribution to total kinetic energy is negligible, because the inertia about that axis of rotation is very small. In this case, the axis of rotation passes through the center of the atoms and mass of an atom is concentrated in the nucleus that is located at the center. If we were to assume that the molecule is rigid (no vibrational energy), the equipartition models says that the internal energy should be

$$u = \frac{5}{2} R T \quad (4.54)$$

for which

$$h = \frac{7}{2} RT, \quad c_v = \frac{5}{2} R \quad c_p = \frac{7}{2} R, \quad \gamma = \frac{7}{5} \quad (4.55)$$

For low-density air, the specific heat ratio γ has been found to be very close to $1.4 = 7/5$, indicating that the equipartition model is valid. However, γ drops below 1.4 as the temperature increases. The equipartition model explains why. As the temperature is increased, the vibrational mode becomes excited, and the molecules can no longer be treated as rigid. When the primary vibrational mode (the “accordion” mode) is fully excited, two more energy modes have been added (one for the vibrational kinetic energy and another for the oscillating potential energy in the intermolecular force field). The equipartition model says that the seven modes should give

$$u = \frac{7}{2} RT, \quad h = \frac{9}{2} RT, \quad c_v = \frac{7}{2} R \quad c_p = \frac{9}{2} R, \quad \gamma = \frac{9}{7} \quad (4.56)$$

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